

Aryan Singh

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Education

Rutgers University, New Brunswick, NJ

Bachelor of Science in Computer Science & Cognitive Science

Expected May 2028

GPA: 3.734 — **Dean's List:** Fall 2025, Spring 2025, Fall 2024

Relevant Coursework: Data Structures, Computer Architecture, Discrete Mathematics, Linear Algebra, Cognition & Decision Making, Linear optimization, Principles of Programming Languages, Minds Machines & Persons

Skills

Languages: Python, JavaScript (Node.js, React.js), Java, HTML/CSS, Scheme

Frameworks/Tools: FastAPI, Flask, React Native, Chrome Extensions, Git/GitHub, Tailwind

AI/ML: HuggingFace Transformers, RAG, TensorFlow.js, scikit-learn, Whisper ASR, ElevenLabs TTS, Computer Vision (CNNs, OCR), Pandas, NumPy, Jupyter, Canvas API, Google Colab

Interests: Soccer, Working-out, Pokémon, Meeting new people, Cars, Photo editing, Music

Experience

Software Engineering Intern

Jan 2026 - Present

HCI Lab for Health & Sustainability, Rutgers University,

New Brunswick, NJ

- Developed a custom ActivityWatch browser watcher to improve classification of user activity, enabling fine-grained tracking of web behavior beyond default labels.
- Built and deployed an interactive environmental data prototype (Next.js, Tailwind, Vercel) to support research on youth engagement with climate threats.
- Translated research requirements into full-stack tools, improving data collection quality and accelerating iterative experimentation.

Data Acquisition

Sept 2024 - Sept 2025

Rutgers Formula Racing,

New Brunswick, NJ

- Evaluated and recommended MoTeC GPS-L10 integration strategy, improving telemetry accuracy by 7% and enabling precise real-time vehicle positioning for competitive lap time optimization and race strategy analysis.
- Designed a comprehensive implementation plan for a CAN bus sensor network enabling real-time monitoring of 15+ vehicle metrics, improving data throughput by 15% and supporting advanced performance analytics.

Projects

NeuroDrive Helper - AI Reading & Accessibility Assistant

Oct 2025 - Present

Python, FastAPI, HuggingFace, RAG, Whisper ASR, ElevenLabs TTS, OCR, Chrome Extension

- Built an AI-powered reading assistant with a RAG pipeline enabling semantic search, summarization, and Q&A across webpages, PDFs, and OCR-captured text, reducing information retrieval time by 40%.
- Developed a Chrome extension integrating OCR, Whisper ASR, and ElevenLabs TTS with accessibility-focused features (text simplification, high-contrast modes, multimodal cues), improving reading speed and comprehension efficiency for neurodiverse users.

FreshCam - AI Fruit Freshness Detection System

Mar 2026

Python, FastAPI, React Native, Computer Vision, Roboflow API, OpenAI API, REST APIs

- Built a full-stack AI system using FastAPI and React Native to classify fruit type and freshness from user-captured images, delivering real-time predictions with sub-2s latency via REST APIs.
- Engineered a hybrid AI pipeline combining a computer vision model (Roboflow) with an OpenAI fallback mechanism and confidence-based switching, reducing prediction failure rates by 30% and improving reliability across varied image conditions.

PokeTrade Assistant - Pokemon Trade Fairness Evaluator (Hackathon Project)

Apr 2025

Flask, React, JavaScript, PokeAPI, TensorFlow.js, CORS, REST APIs

- Built a full-stack Pokemon trade evaluator using Flask and PokeAPI to analyze evolution chains, base stats, levels, and progression data, generating real-time fairness scores for any two Pokemon combinations.
- Developed a React frontend with interactive input forms and dynamic result rendering, integrating REST endpoints from the Flask backend to deliver instant trade decisions and improve user decision accuracy during testing.